

WOLF POINT 34 NE, MT, USA

WMO: 742310

Lat: 48.4887N

Lon: 105.2096W

Elev: 806

StdP: 92.01

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 94059

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-27.6	-25.0	-30.7	0.2	-27.2	-28.2	0.3	-24.8	12.6	-5.1	11.0	-9.3	4.1	N/A	0.657

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	32.4	17.2	30.4	17.0	28.5	16.7	19.9	27.0	18.6	26.3	17.6	25.6	4.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.1	13.5	24.0	15.8	12.4	22.2	14.7	11.5	20.9	60.4	27.2	55.8	26.3	52.6	25.5	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.1	9.9	8.8	-30.8	35.9	1.7	2.2	-32.0	37.5	-33.0	38.8	-33.9	40.0	-35.1	41.7		
(5)			-30.8	22.4	1.3	1.9	-31.7	23.7	-32.4	24.8	-33.1	25.9	-34.1	27.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.5	-9.1	-9.5	-1.7	5.6	11.4	16.6	20.9	19.7	14.5	6.2	-1.6	-8.1	(6)
(7)		DBStd	12.50	8.50	7.72	7.88	5.72	5.09	3.93	3.55	4.11	5.38	6.05	7.03	7.82	(7)
(8)		HDD10.0	2778	592	546	367	151	44	2	0	0	18	146	349	563	(8)
(9)		HDD18.3	4913	850	779	623	382	220	78	14	33	139	376	597	821	(9)
(10)		CDD10.0	1134	0	0	3	19	88	199	338	302	153	30	2	0	(10)
(11)		CDD18.3	226	0	0	0	0	5	26	94	76	24	1	0	0	(11)
(12)		CDH23.3	2337	0	0	0	7	69	234	951	799	261	15	0	0	(12)
(13)		CDH26.7	835	0	0	0	1	11	57	367	310	88	2	0	0	(13)
(14)	Wind	WSAvg	4.3	4.4	4.2	4.8	5.0	4.8	4.1	3.9	4.0	4.1	4.4	4.2	4.0	(14)
(15)	Precipitation	PrecAvg	349	11	7	13	25	54	76	55	41	24	22	11	10	(15)
(16)		PrecMax	517	30	18	41	55	150	155	203	158	64	78	43	22	(16)
(17)		PrecMin	222	3	2	2	9	16	24	16	1	6	3	1	1	(17)
(18)		PrecStd	61	7	4	8	13	35	30	36	32	17	18	9	5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	8.9	19.3	24.2	28.2	31.3	35.5	35.5	32.6	25.5	17.0	8.5	(19)
(20)			MCWB	3.3	3.9	8.9	11.7	14.5	18.3	16.8	16.9	16.8	13.7	8.8	3.2	(20)
(21)		2%	DB	5.3	5.4	14.6	20.2	25.3	28.3	32.6	32.5	29.1	21.4	12.9	5.7	(21)
(22)			MCWB	2.0	1.8	6.9	9.5	13.5	17.1	17.9	16.6	15.2	11.6	6.8	1.9	(22)
(23)		5%	DB	3.6	2.8	11.4	17.4	22.9	25.9	30.6	30.1	26.5	17.9	10.1	3.7	(23)
(24)			MCWB	0.9	0.4	5.1	8.1	12.2	15.9	17.8	16.3	14.2	9.9	5.0	0.7	(24)
(25)		10%	DB	1.8	0.8	8.4	14.9	20.4	24.0	28.6	27.9	23.5	15.1	7.5	1.7	(25)
(26)			MCWB	-0.3	-0.9	3.7	6.7	11.1	14.8	17.8	16.2	13.1	8.3	3.5	-0.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.0	4.6	9.0	12.1	16.8	21.2	22.6	20.1	18.8	14.4	9.3	3.9	(27)
(28)			MCDB	8.3	9.2	17.9	23.4	23.8	27.5	28.7	27.9	27.5	23.7	17.3	7.4	(28)
(29)		2%	WB	2.1	2.4	7.6	9.7	15.0	18.3	20.5	18.7	16.2	12.5	7.1	2.1	(29)
(30)			MCDB	5.0	5.8	15.0	18.2	22.3	25.6	27.9	27.2	25.7	19.2	12.8	5.6	(30)
(31)		5%	WB	0.9	0.8	5.8	8.0	13.6	16.8	19.2	17.8	14.9	10.3	5.2	0.8	(31)
(32)			MCDB	3.2	3.0	11.3	15.6	21.0	23.9	26.3	26.4	23.3	16.5	9.7	3.6	(32)
(33)		10%	WB	-0.2	-0.5	4.1	6.7	12.3	15.7	18.1	17.0	13.7	8.8	3.4	-0.5	(33)
(34)			MCDB	1.8	1.3	8.5	13.5	19.0	22.4	25.6	25.4	22.1	14.7	6.9	1.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.3	8.3	9.4	11.0	11.2	10.6	12.2	12.3	11.5	9.9	8.7	7.5	(35)
(36)			MCDBR	8.7	10.0	13.4	14.8	14.1	13.1	14.7	14.8	14.9	14.1	11.6	8.9	(36)
(37)			MCWBR	6.6	7.6	7.6	7.5	6.5	6.3	5.5	5.2	6.0	6.8	6.8	6.1	(37)
(38)		5% WB	MCDBR	8.8	9.9	13.4	14.1	12.3	11.6	12.7	13.2	13.4	12.8	11.6	8.8	(38)
(39)			MCWBR	6.7	7.5	7.9	7.5	6.1	6.4	6.3	5.2	5.9	6.4	7.0	6.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.246	0.262	0.270	0.314	0.332	0.344	0.352	0.359	0.316	0.281	0.253	0.237	(40)	
(41)		taud	2.465	2.462	2.548	2.430	2.430	2.440	2.435	2.384	2.473	2.564	2.535	2.499	(41)	
(42)		Ebn at Noon	856	916	963	939	929	916	903	880	893	882	843	827	(42)	
(43)		Edh at Noon	60	77	84	105	109	109	108	109	90	69	56	51	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1.45	2.48	3.74	4.94	5.88	6.38	6.85	5.60	4.11	2.60	1.56	1.10	(44)	
(45)		RadStd	0.13	0.16	0.23	0.22	0.46	0.37	0.33	0.39	0.38	0.26	0.17	0.06	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page